CertiChain

Blockchain-Based Academic Certificate Verification System

Project Type: Blockchain / Web Application
Purpose: Secure academic certificate issuance, tamper detection, and public verification.

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Problem Statement

Traditional academic certificates are primarily dependent on centralized institutional records and physical or digitally generated documents. Verification of a certificate often requires manual communication with the issuing institution, which can make the process time-consuming and difficult to scale.

Digitally generated certificates can also be modified after issuance. A verifier receiving a certificate needs a reliable mechanism to determine whether the certificate was genuinely issued by the institution and whether the document has been altered.

Proposed Solution

CertiChain is a blockchain-enabled academic certificate management and verification platform.
The system combines:

- A web-based institutional certificate management platform
- PostgreSQL for structured institutional and certificate data
- A blockchain smart contract for certificate registration and issuer authorization
- Cryptographic hashing for document integrity verification
- Wallet-based institutional identity
- Role-based access control
- Multi-institution support
- Public certificate verification

The certificate lifecycle



The blockchain does not store the complete PDF certificate. Instead, the certificate's cryptographic fingerprint and relevant blockchain information are registered on-chain. This provides an integrity mechanism while keeping the actual document in the application's storage layer.

Blockchain Technologies Used

Solidity:

Solidity is used to develop the CertiChain.sol smart contract containing the core on-chain rules for issuer authorization, certificate registration, revocation, and certificate state.

Ethereum / EVM:

The Ethereum Virtual Machine (EVM) provides the execution environment for the Solidity smart contract. CertiChain uses this model so that certificate-related blockchain operations are executed according to transparent and predefined smart-contract rules.

Hardhat:

Hardhat is used as the blockchain development and testing framework. It provides the local EVM network used to deploy and test the CertiChain smart contract, including issuer authorization and certificate transaction workflows.

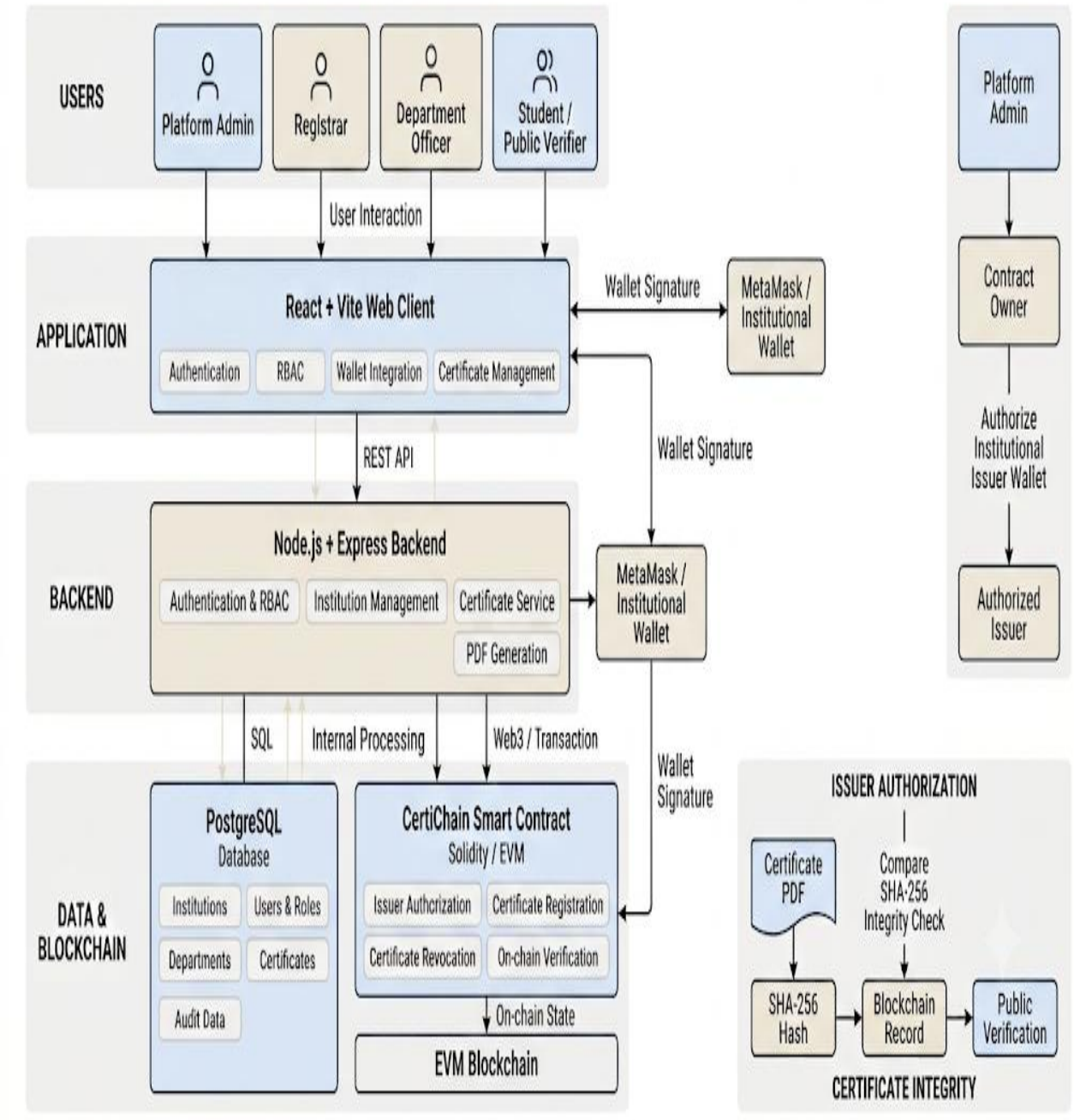
OpenZeppelin Ownable:

OpenZeppelin's Ownable mechanism is used to control privileged smart-contract operations. In CertiChain, the contract owner is responsible for authorizing institutional issuer wallets before they can perform permitted certificate operations.

MetaMask:

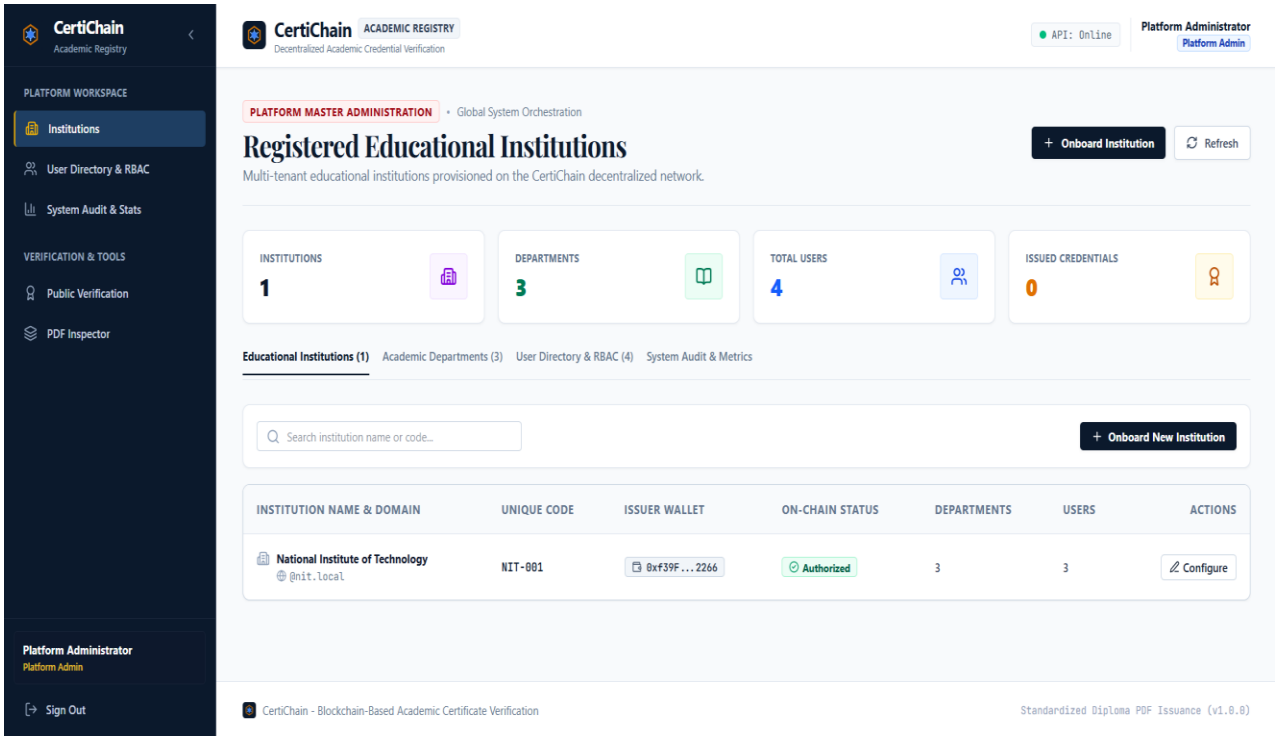
MetaMask acts as the user's blockchain wallet and transaction-signing interface. Institutional wallets are connected to CertiChain through MetaMask, allowing authorized issuers to sign blockchain transactions without exposing their private keys to the application.

Architecture

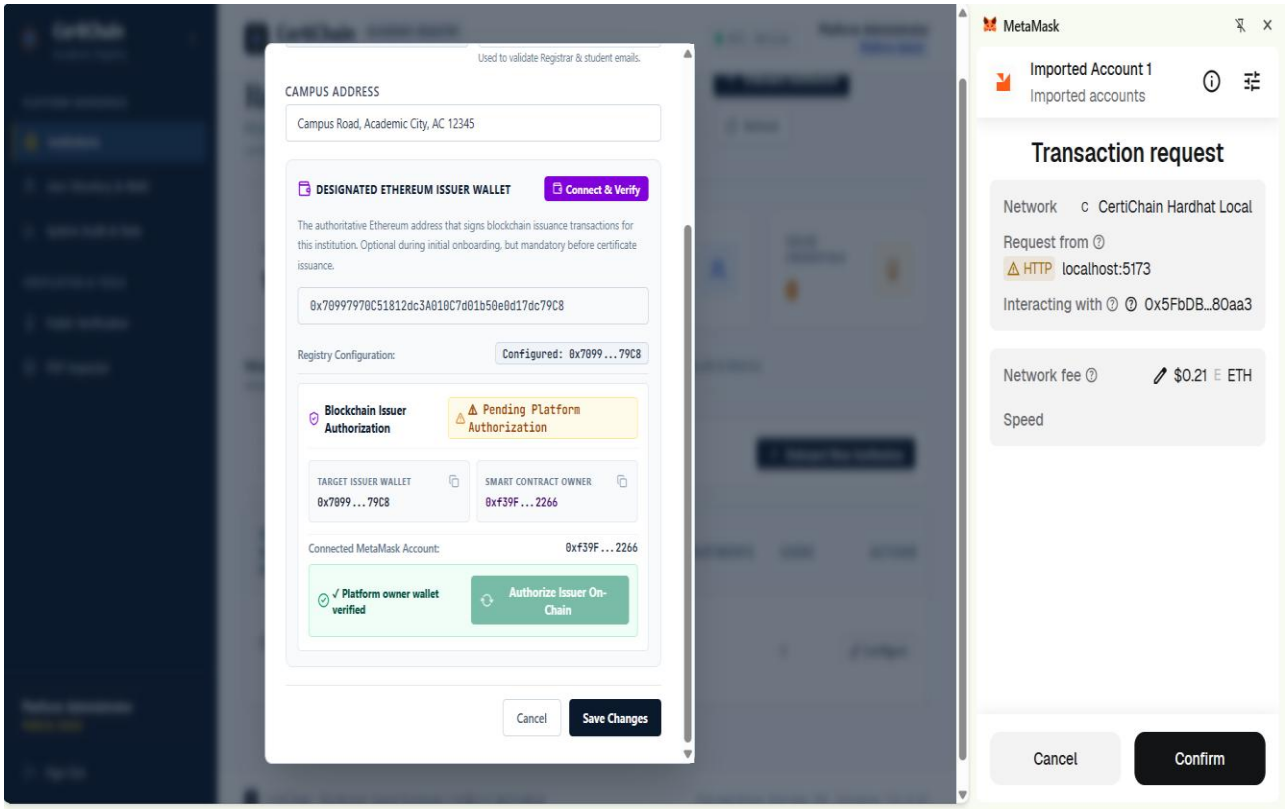


Project Output

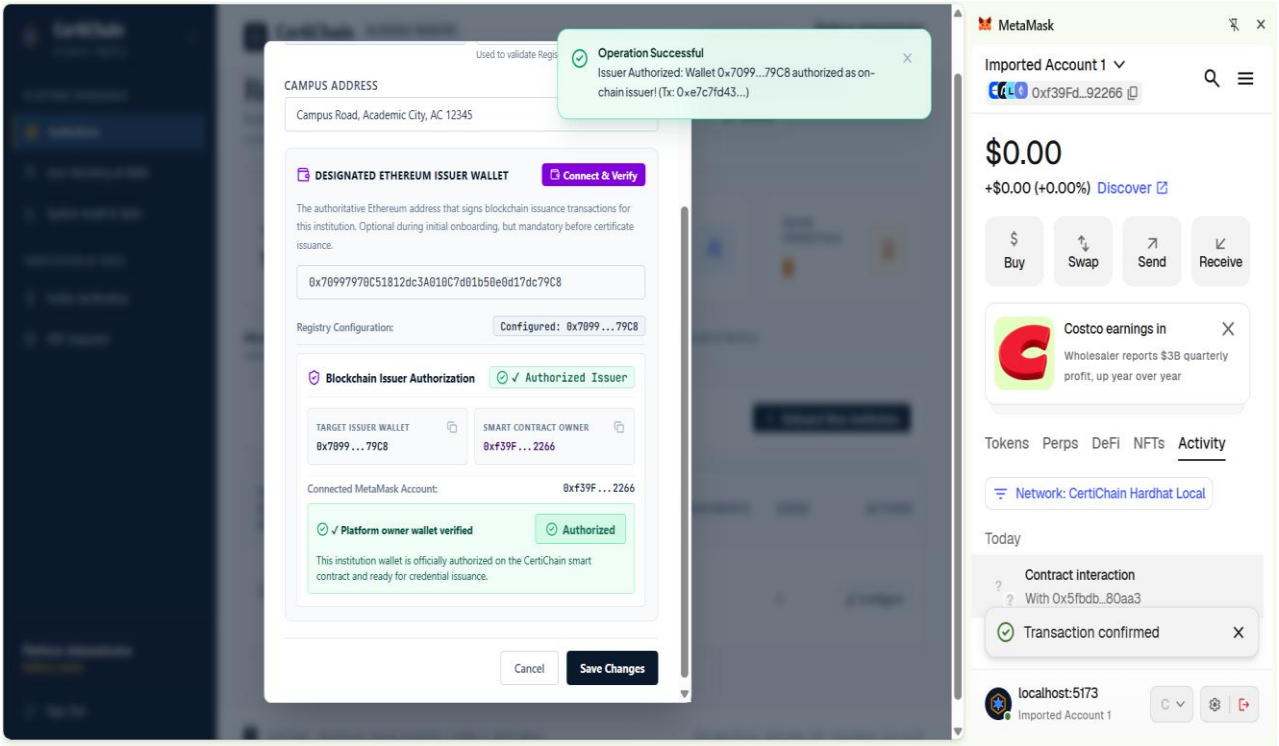
Admin user dashboard:



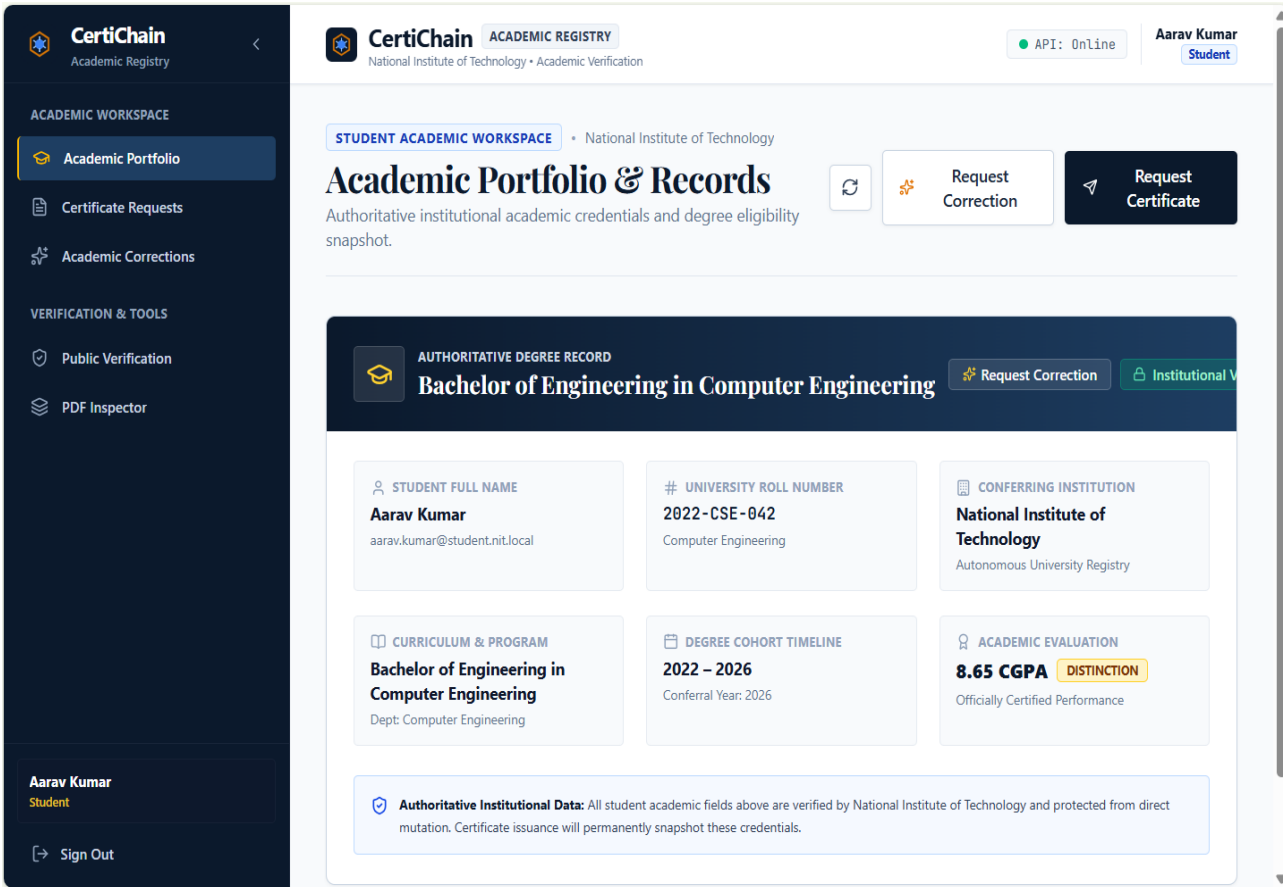
Admin authorizing institute for certificate issuance



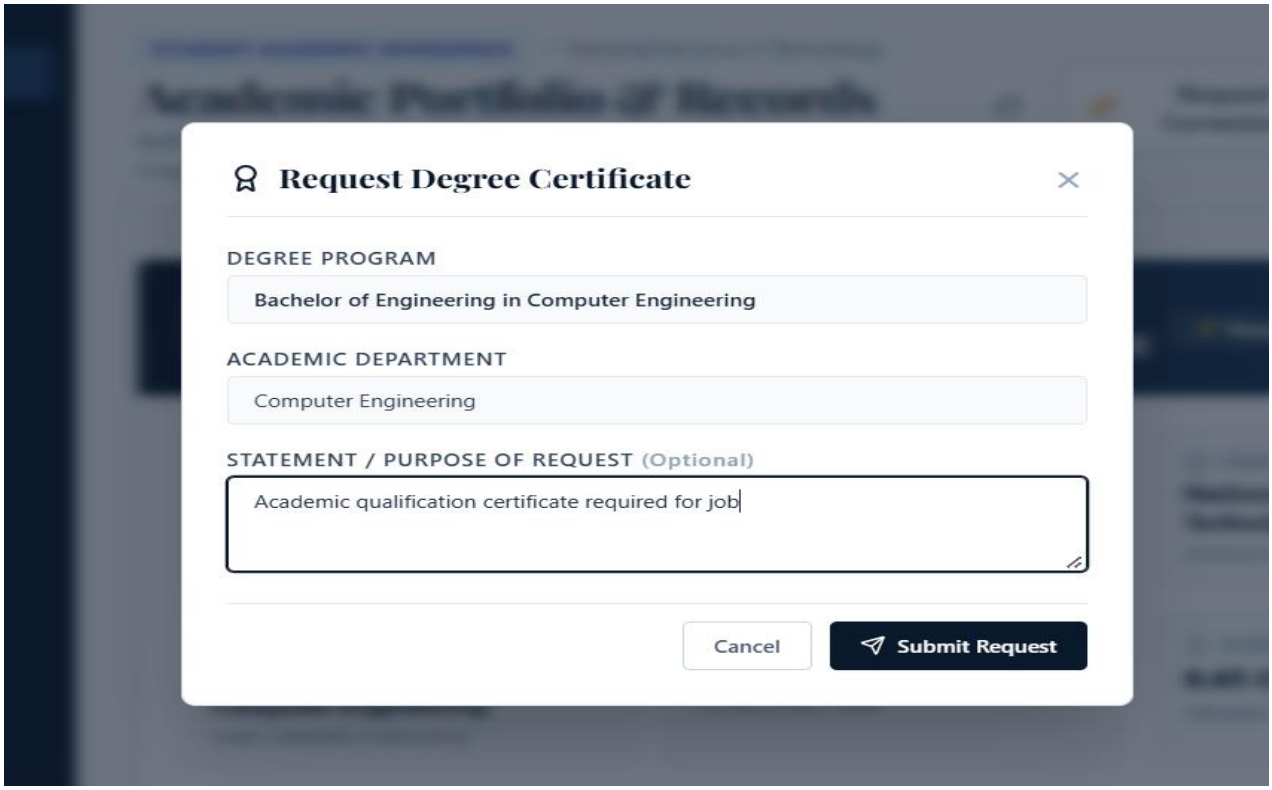
Institute wallet address is authorized using admin wallet



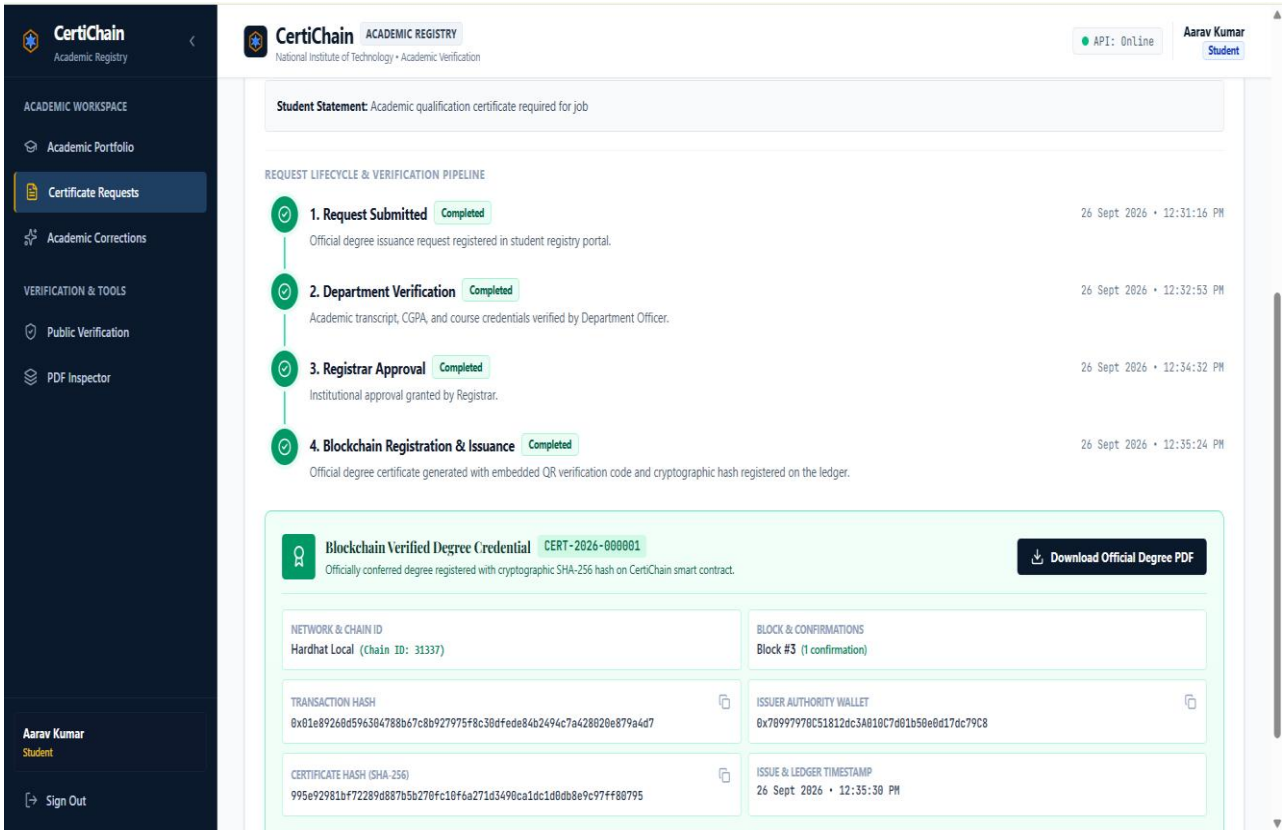
Student dashboard



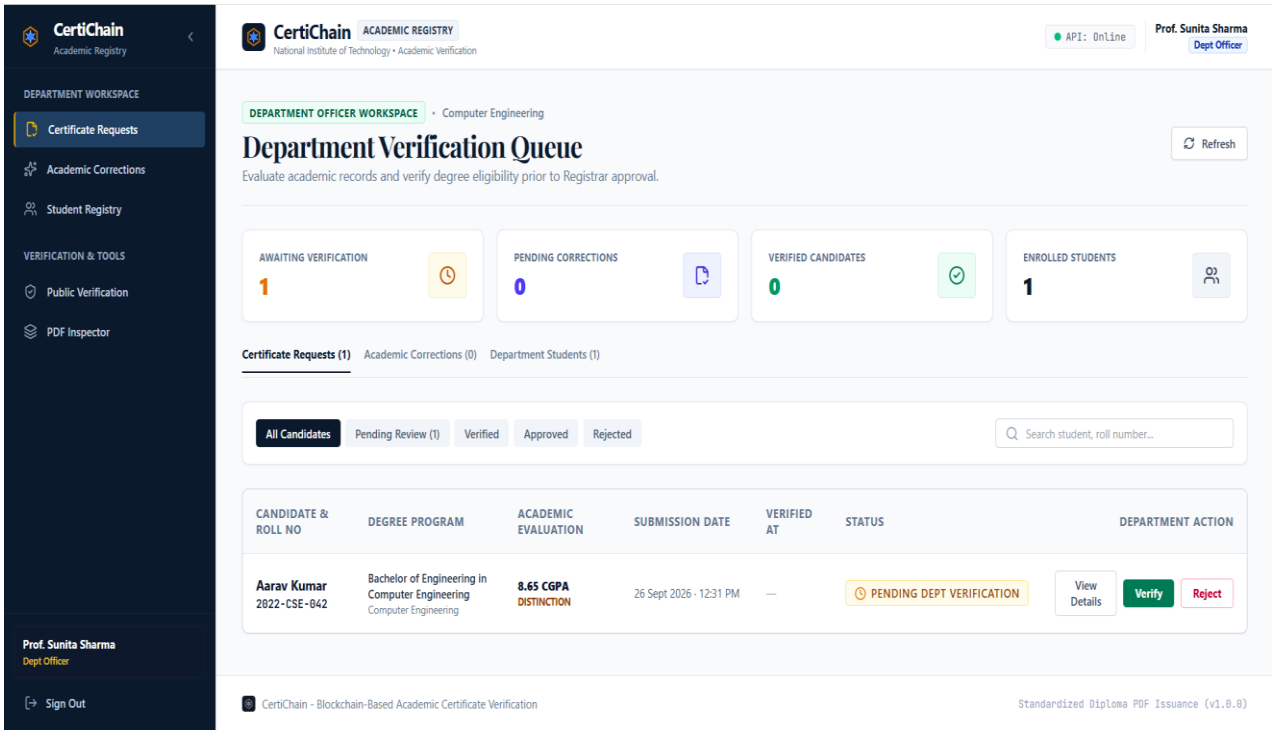
Student requests for the certificate



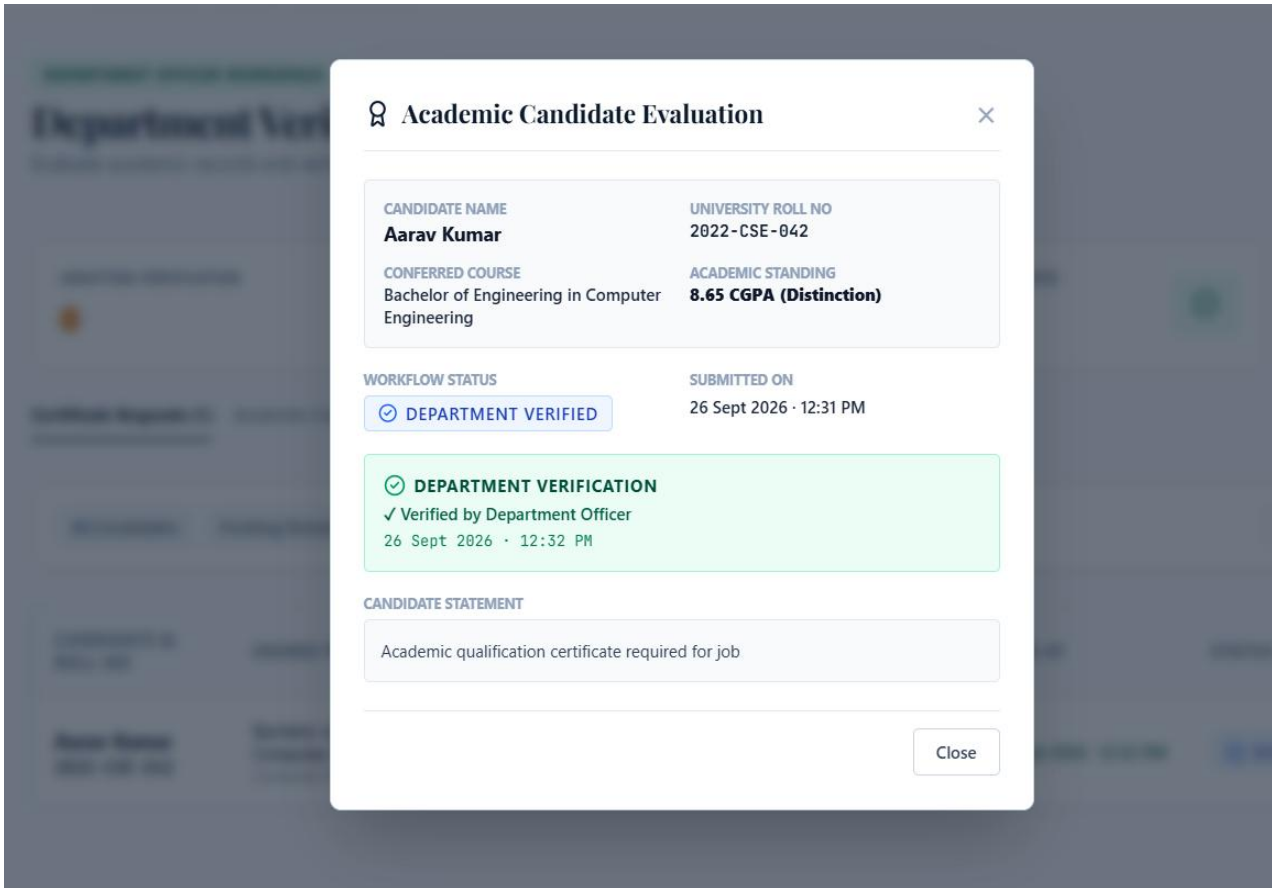
Certificate issued and details of blockchain registration



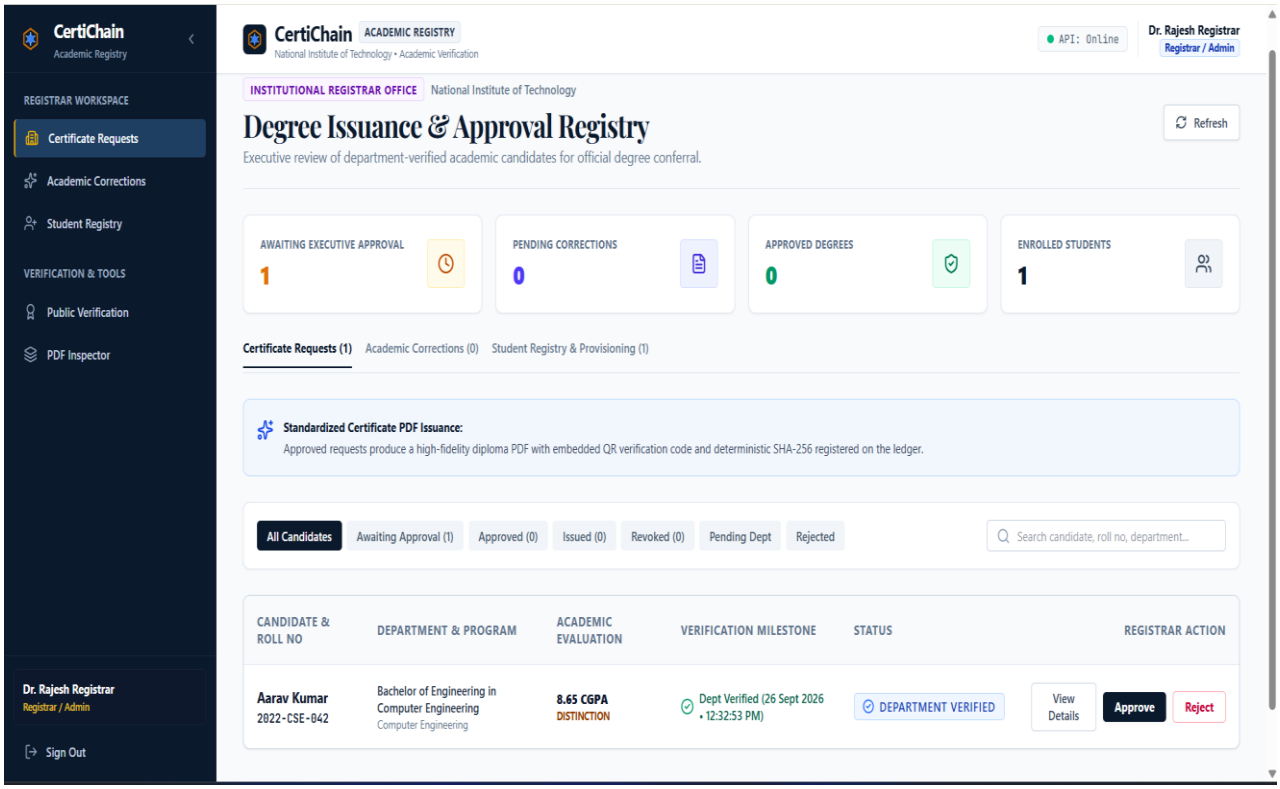
Department officer dashboard



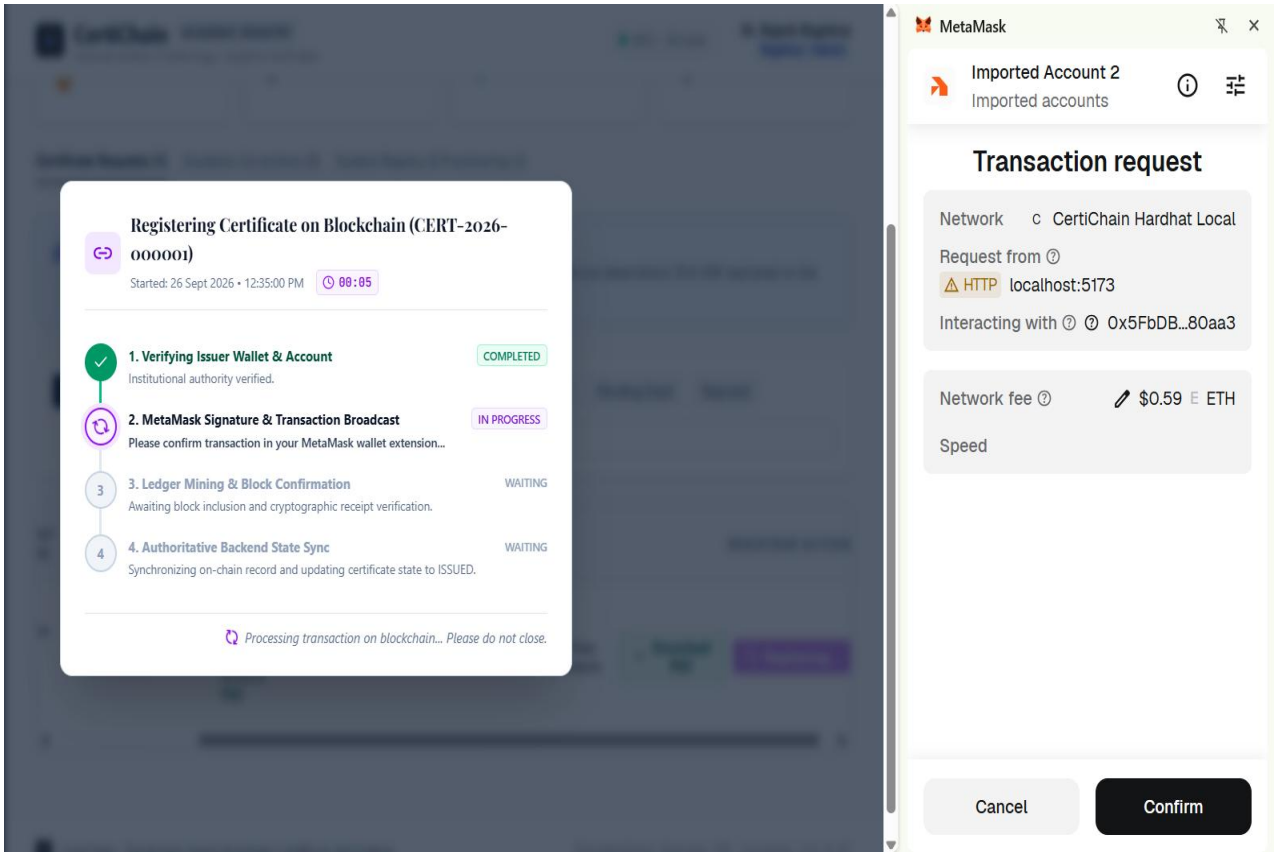
Department officer verifies the certificate request



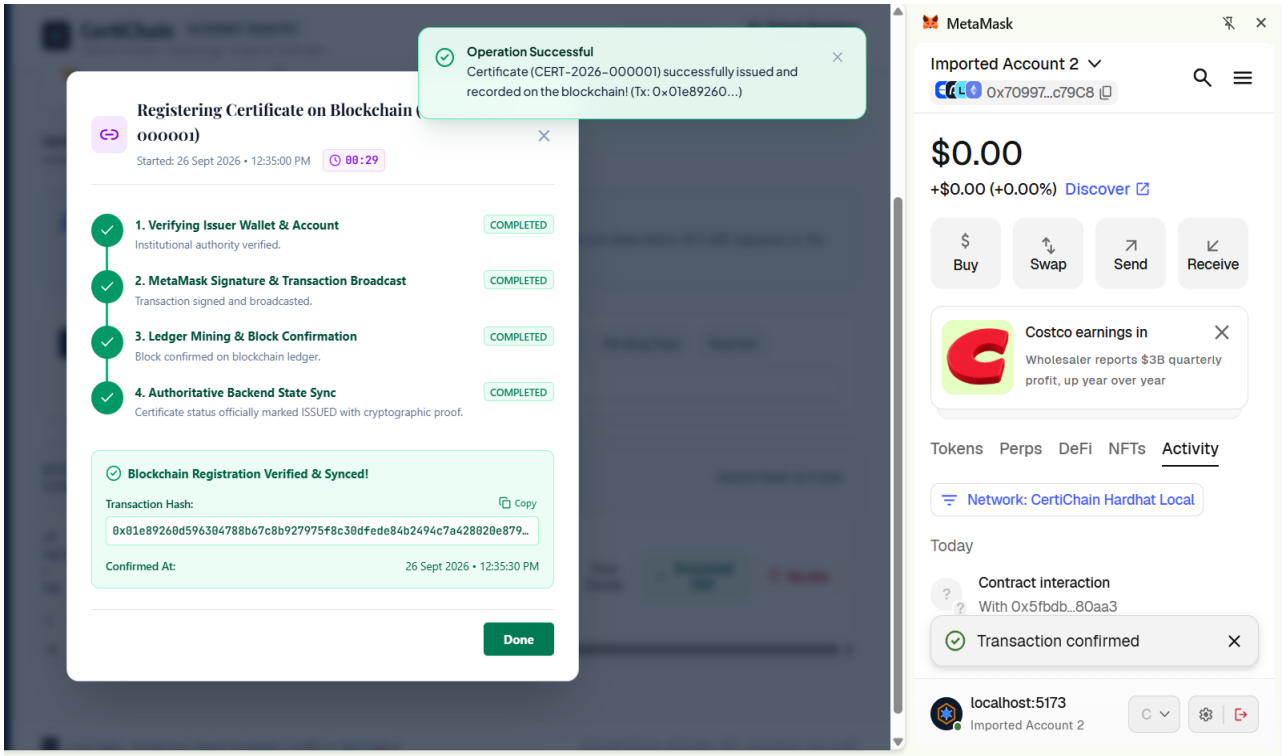
Registrar dashboard



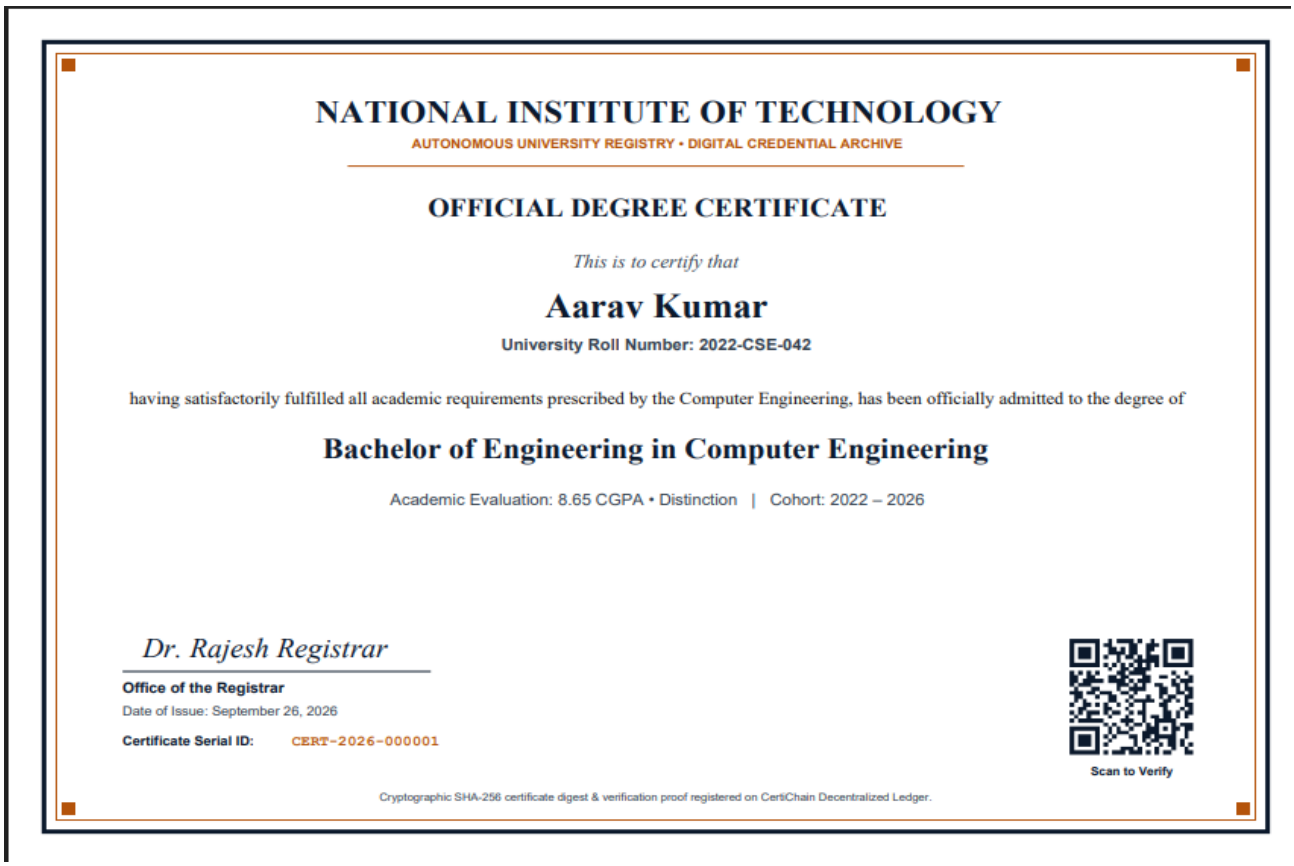
Certificate registration on blockchain



Certificate registered by the registrar of the institute



Certificate issued



NATIONAL INSTITUTE OF TECHNOLOGY

AUTONOMOUS UNIVERSITY REGISTRY • DIGITAL CREDENTIAL ARCHIVE

OFFICIAL DEGREE CERTIFICATE

This is to certify that

Aarav Kumar

University Roll Number: 2022-CSE-042

having satisfactorily fulfilled all academic requirements prescribed by the Computer Engineering, has been officially admitted to the degree of

Bachelor of Engineering in Computer Engineering

Academic Evaluation: 8.65 CGPA • Distinction | Cohort: 2022 – 2026

Dr. Rajesh Registrar

Office of the Registrar

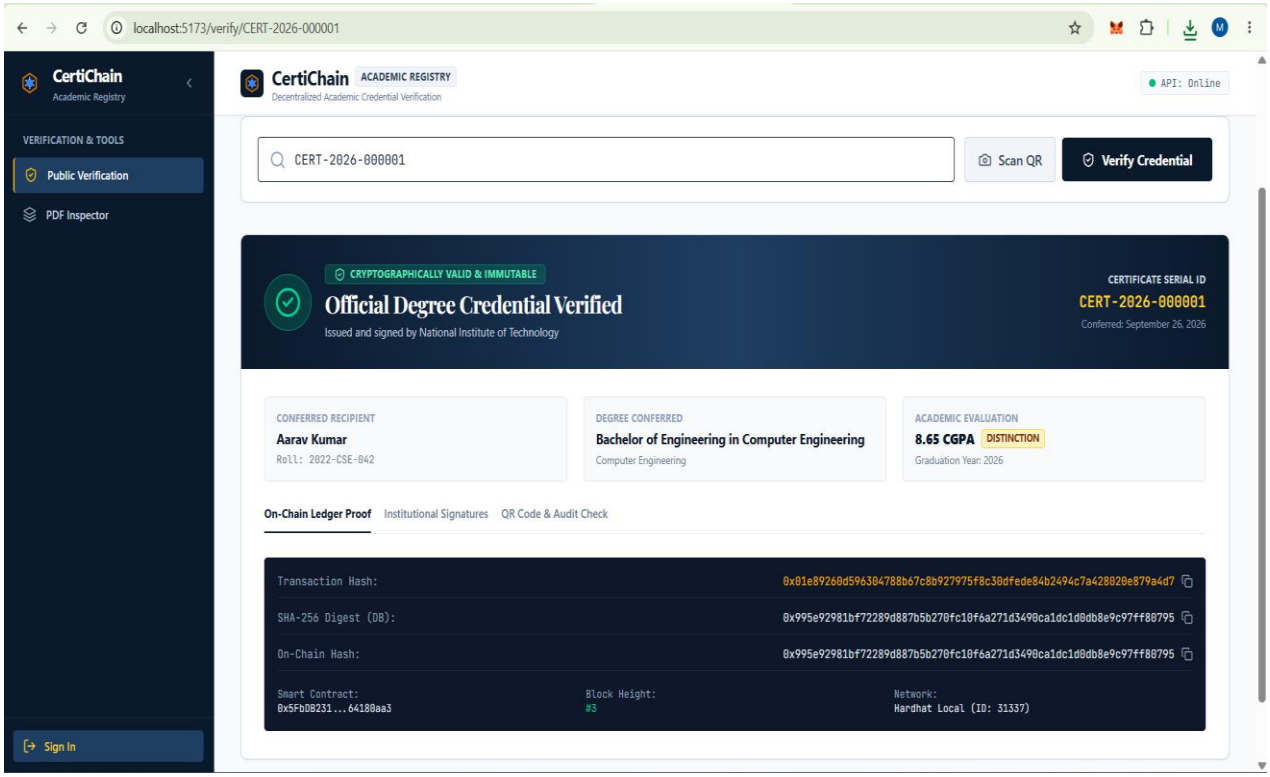
Date of Issue: September 26, 2026

Certificate Serial ID: CERT-2026-000001

Scan to Verify

Cryptographic SHA-256 certificate digest & verification proof registered on CertiChain Decentralized Ledger.

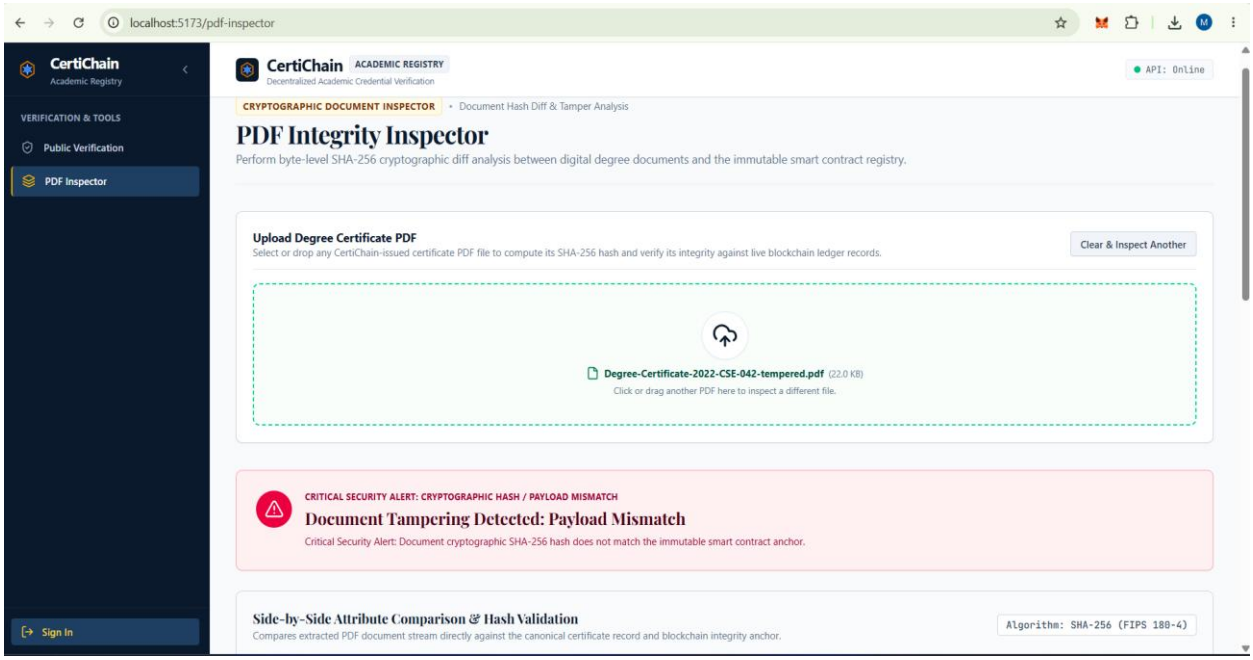
Public verification portal



Tempered certificate details



PDF integrity checking against blockchain registered certificate



PDF tempering detection

